

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

Group Project Report

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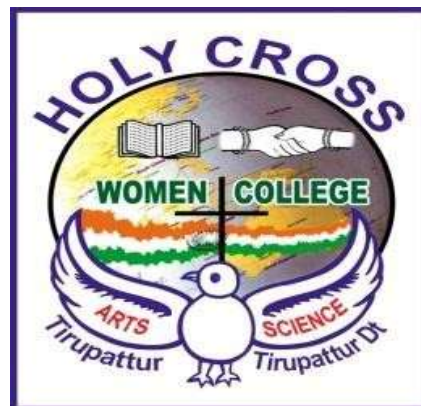
*In Partial Fulfilment for The Award of The Degree
Of*

BACHLER OF COMPUTER SCIENCE

Under the supervision of

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352, HOLY CROSS ARTS AND SCIENCE COLLEGE FOR WOMEN

(Affiliated to Thiruvalluvar University)

DEPARTMENT OF COMPUTER SCIENCE

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Student Management & Course Management System (SECMS)

User Story:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting within a unified Salesforce environment.

1. INTRODUCTION

1.1 Project Overview:

The **Student Enrollment & Course Management System (SECMS)** is a centralized web-based application designed to automate and streamline the core academic processes of an educational institution. It provides a unified platform for managing student admissions, enrollment, course creation, course registration, attendance, and grade management.

- ☐ Reduce administrative workload and processing time.
- ☐ Minimize data duplication and human errors.
- ☐ Improve data accuracy, consistency, and security.
- ☐ Provide real-time access to academic information for all stakeholders.

The project will be developed using a structured software engineering approach, covering ideation, requirement analysis, system design, planning, and implementation phases, resulting in a scalable and secure solution suitable for colleges and universities.

1.2 Purpose:

The purpose of the Student Enrollment & Course Management System (SECMS) is to provide a single, integrated digital platform that automates and simplifies the key academic and administrative processes related to student enrollment and course management.

- Automate enrollment workflows
- Centralize course and academic data
- Improve efficiency and reduce errors
- Enhance transparency and accessibility
- Support decision-making and reporting
- Ensure security and data integrity

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Question	Answer
Customer	Main Store and Sub Store teams
Cause	Manual and disconnected processes
Impact	Delays, errors, and lack of tracking
Need	Centralized and automated logistics system
Solution	My Logistics – Main Store to Sub Store Management System

1. IDEATION PHASE

1.1 Problem Statement (Customer Problem Statement Template)

S.No	Customer Problem Statement	Description
1	 Who is the customer?	Customers are users who need to transfer materials between the Main Store and Sub Store.
2	 What is the problem?	The material request and transfer process is difficult to manage manually.
3	 Where does the problem occur?	The problem occurs during material request, approval, packing, dispatch, transportation and receiving.
4	 Why is it a problem?	Manual processes can cause delays, errors, missing information and difficulty in tracking materials.
5	 How does it affect the customer?	Customers may not know the current status of their material request or delivery.
6	 Current method used	Requests and material movements may be managed using manual records, phone calls, messages or spreadsheets.
7	 Customer need	Customers need a simple system to raise requests and track material movement in real time.
8	 Expected solution	A centralized logistics management system that connects the Sub Store, Main Store Manager, Packing, Dispatch, Transport and Receiving teams.
9	 Expected benefit	Faster processing, better tracking, reduced errors and improved communication between teams.
10	 Final Problem Statement	There is a need for a centralized logistics management system to efficiently manage and track material transfer between the Main Store and Sub Store from request to delivery.

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- ☐ Reduce material transfer delays
- ☐ Provide real-time tracking
- ☐ Reduce manual errors
- ☐ Improve communication between teams
- ☐ Provide accurate material information
- ☐ Centralize the entire logistics process
- ☐ Make the system simple and user-friendly

1.1 Empathy Map Canvas



2.3 Brainstorming & Idea Prioritization

A. Brainstorming

- Identify the Problem – Clearly understand the logistics problem.
- Generate Ideas – Encourage team members to suggest different solutions.
- Discuss & Combine – Discuss ideas and combine similar ideas.
- Shortlist Ideas – Select practical and useful ideas.

B. Idea Prioritization

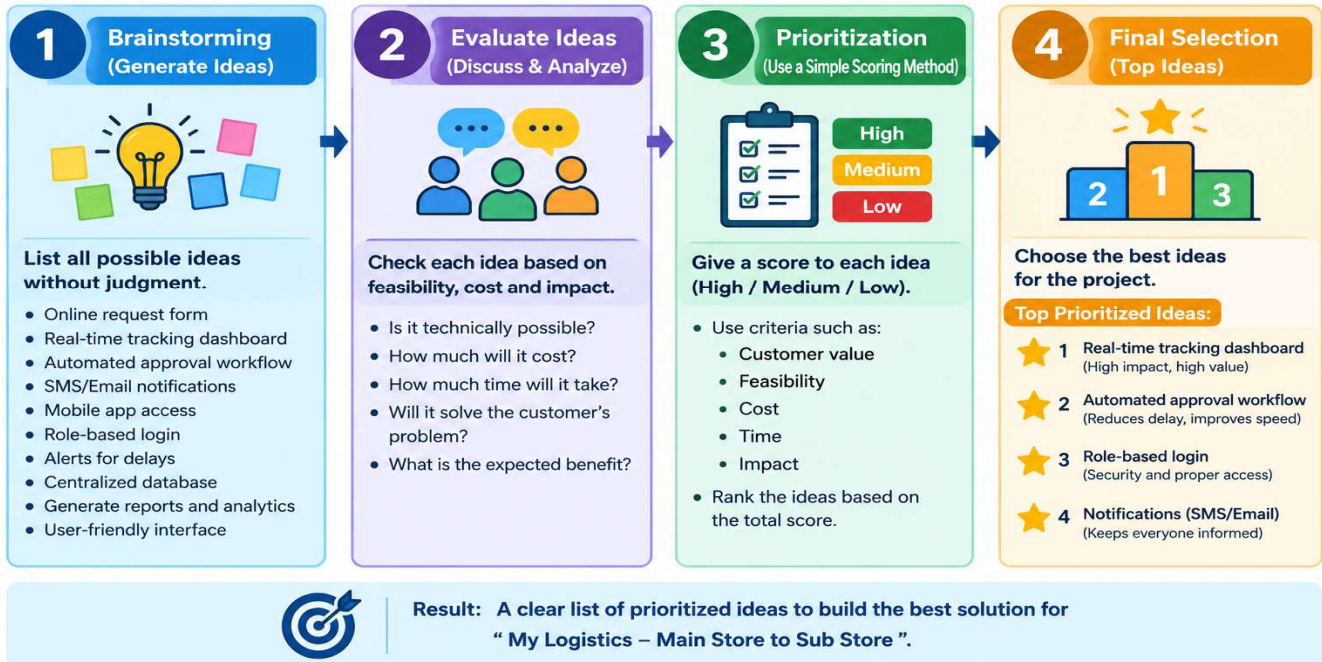
- Define Criteria – Consider impact, feasibility, cost, and effort.
- Score Ideas – Give each idea a score based on the criteria.
- Rank Ideas – Compare scores and arrange ideas by priority.
- Select & Plan – Select high-priority ideas for implementation.

Simple Flow

Problem Identification → Idea Generation → Discussion → Shortlisting → Scoring → Ranking → Final Selection.

2.3 Brainstorming & Idea Prioritization

Generate ideas → Evaluate → Prioritize → Select the best solution

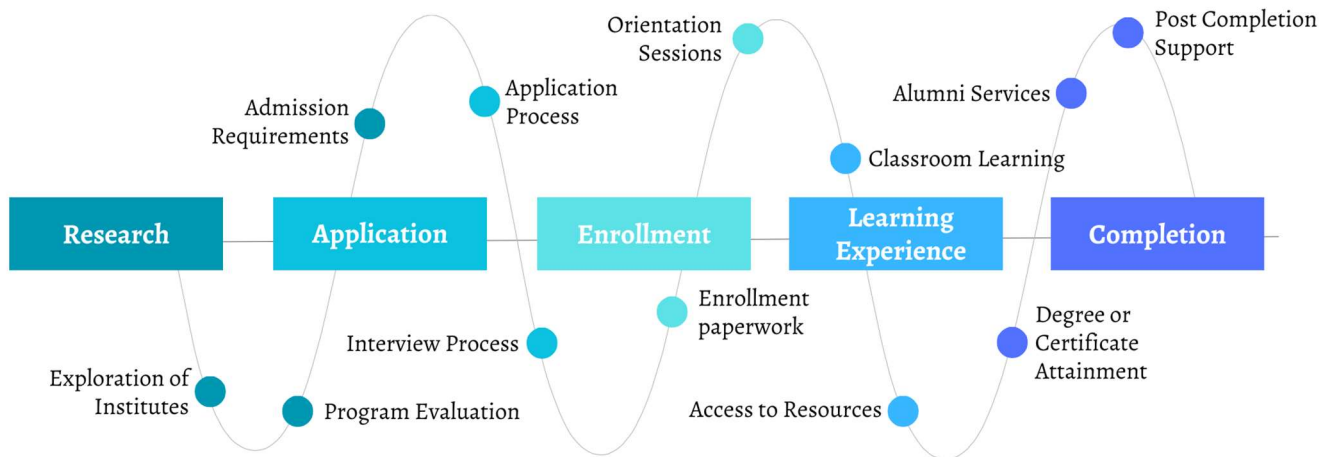


3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

- ☐ Online material request
- ☐ Role-based login
- ☐ Request approval
- ☐ Packing and dispatch management
- ☐ Transport tracking
- ☐ Real-time status updates
- ☐ Receiving confirmation
- ☐ Notification system
- ☐ Centralized database
- ☐ Reports and request history

Customer Journey Map for Education



Non-Functional Requirements

NFR-1: Performance

Meaning: The system should respond quickly when users perform an action.

Example:

When a student logs in or opens the course list, the page should load within a few seconds.

NFR-2: Security

Meaning: Student and course information must be protected from unauthorized access.

Example:

Only the Admin can delete student records, while students can only view and update their own information.

NFR-3: Reliability

Meaning: The system should work correctly and consistently without frequent failures.

Example:

When a student registers for a course, the registration information should be saved correctly.

NFR-4: Usability

Meaning: The system should be simple and easy to use.

Example:

Students should easily find options such as Login, My Courses, Profile, Results, and Logout.

NFR-5: Scalability

Meaning: The system should handle an increasing number of students and courses.

Example:

The system should continue working efficiently when the number of students increases from **500 to 5,000**.

NFR-6: Availability

Meaning: The system should be available whenever users need it.

Example:

Students should be able to access their course details and learning materials at any time, except during.

NFR-7: Backup and Recovery

Meaning: The system should regularly back up important data and recover it after a failure.

Example:

If the database crashes, student records and course information can be restored from the latest.

NFR-8: Compatibility

Meaning: The system should work properly on different devices, browsers, and operating systems.

Example:

Students should be able to use the system on Chrome, Edge, Firefox, mobile phones, tablet.

3.2 Data Flow Diagram

1. External Entities (Sources & Destinations)

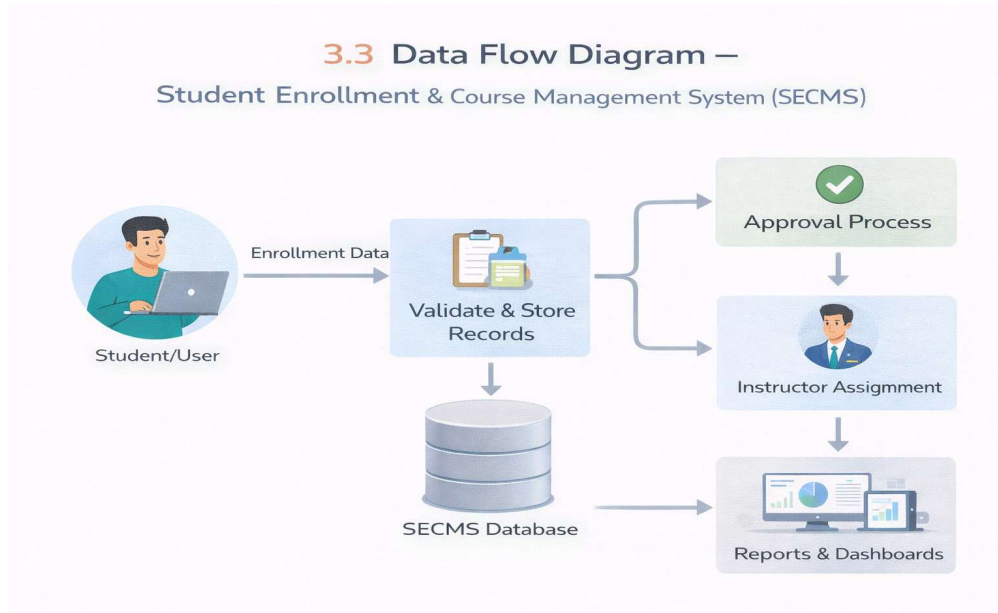
- **Student:** Submits registration, course selection, and views marks/schedules.
- **Faculty:** Submits grades, course materials, and manages attendance.
- **Admin:** Manages course offerings, fee updates, and overall user permissions.

2. DFD Abstraction Levels

Level 0 (Context Diagram): High-level view showing the entire system as a single process node connected to external entities.

Level 1: Breaks the single context process into major sub-processes, primary data stores, and core data flows.

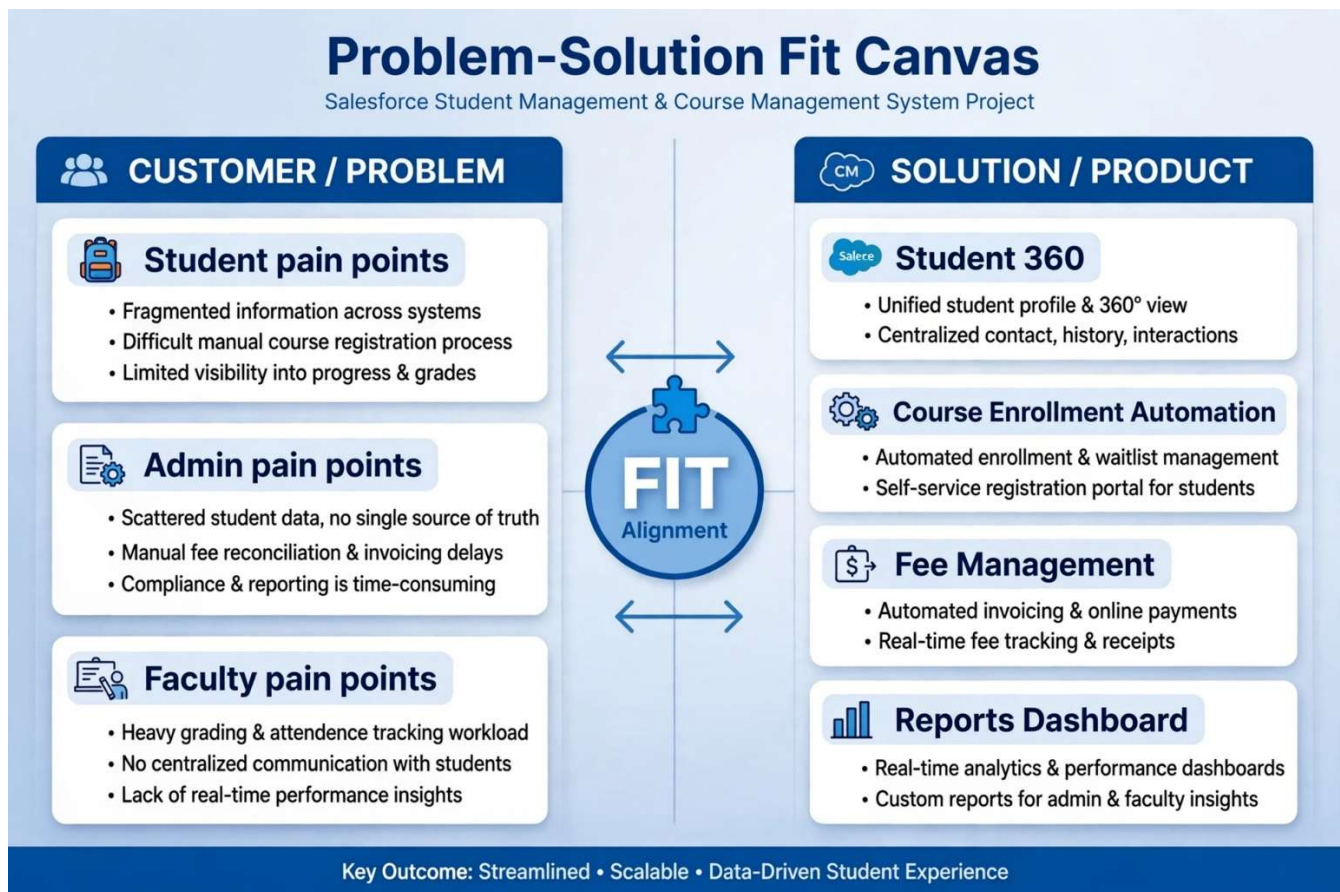
Level 2 (Detailed DFD): Decomposes Level 1 sub-processes further into lower-level functional steps.



4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The purpose of the Problem-Solution Fit is to validate that the proposed Salesforce-based system solves the real pain points of students, admins, and faculty.



4.2 Proposed Solution

The proposed system is a cloud-based Student and Course Management System built on Salesforce Platform to replace manual Excel/Paper-based processes.

Core Objectives:

- Centralize all student data into a single Student 360 view
- Automate course enrolment, fee collection, and grading
- Provide self-service portals for students and faculty
- Enable real-time reporting for management

Key Features Proposed:

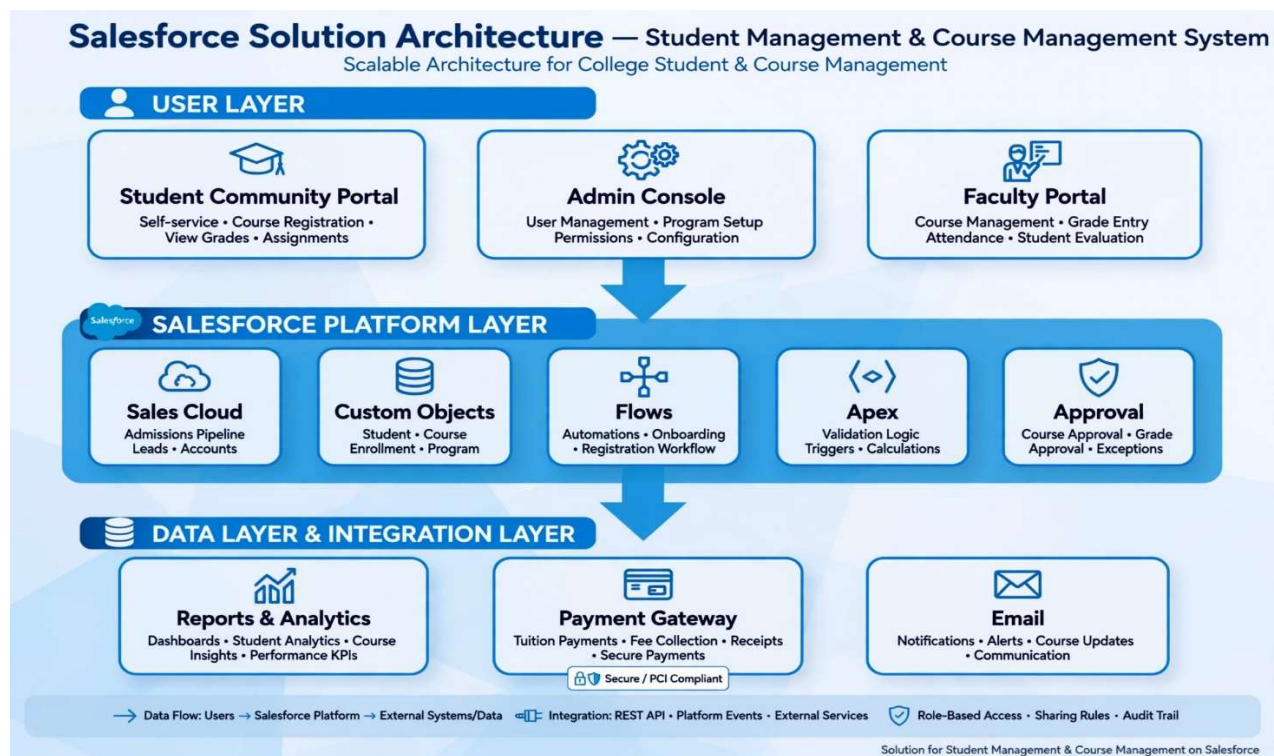
1. Student Management Module:

- Custom Object: Student (fields: Roll No, Contact, Department, Year, Status)
- Auto Student ID generation via Flow
- Duplicate prevention with Matching Rules
- Student lifecycle: Applicant > Enrolled > Alumni

2. Course Management Module:

- Custom Objects: Course, Program, Enrolment
- Fields: Course Code, Credits, Seats, Faculty, Prerequisites
- Validation: Seat availability check, Prerequisite check
- Automated Waitlist management

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

This is the complete list of User Stories for Salesforce Student & Course Management System.

Epic	User Story ID	User Story	Priority	Story Points	Acceptance Criteria
EPIC 1: Student Management	US-01	As an Admin, I want to create a student record with auto ID generation	High	5	Auto ID, Duplicate check, Required fields validation
	US-02	As an Admin, I want to view Student 360 with all enrolments & fees	High	8	Related lists, 360 view pages
	US-03	As a Student, I want to login to portal and view my profile	High	5	Experience Cloud Login
EPIC 2: Course Management	US-04	As an Admin, I want to create Courses with seat limit & prerequisites	High	5	Validation for seats, prereq check
	US-05	As a Student, I want to enrol in courses via self-service portal	High	13	Flow, Seat availability, Waitlist
	US-06	As a System, I want to auto-update seat count on enrolment	Medium	3	Apex Trigger
EPIC 3: Fee Management	US-07	As an Admin, I want to generate fee invoices automatically	High	8	Auto invoice on enrollment
	US-08	As a Student, I want to pay fees online & get receipt	High	8	Payment gateway integration, Email receipt

SPRINT SCHEDULE:

We divided the 6 weeks into 4 Sprints.

Sprint	Duration	Goal	User Stories Included	Deliverable
Sprint 1: Foundation	Week 1 - 1.5	Org Setup & Data Model	US-01, US-04, US-12	Dev Org, Custom Objects: Student, Course, Enrolment, Fee, Roles & Profiles
Sprint 2: Core Development	Week 2 – 3	Automation & Business Logic	US-02, US-06, US-07, US-10	Flows, Apex Triggers (Seat count, GPA), Validation Rules, Approval Process
Sprint 3: Portal & Integration	Week 4 - 5	Community Portal & Payment	US-03, US-05, US-08, US-09	Experience Cloud Student Portal, Screen Flow for enrolment, Email Alerts
Sprint 4: Testing & Deployment	Week 5.5 - 6	Testing, Reports & Go-Live	US-11 + Testing	Test Classes (>75% coverage), Dashboards, UAT, Deployment to Production

6. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation

Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

Milestone 3: Tabs Creation

Milestone 4: Lightning App Creation

Milestone 5: Fields Creation

Milestone 6: Create Relationships

Milestone 7: Page Layout Customization

Milestone 8: Validation Rules

Milestone 9: Flows (Automation)

Milestone 10: Approval Process

Milestone 11: Reports

Milestone 12: Dashboards

Milestone 13: Security Setup

Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Develop reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

Milestone 1-Salesforce:

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

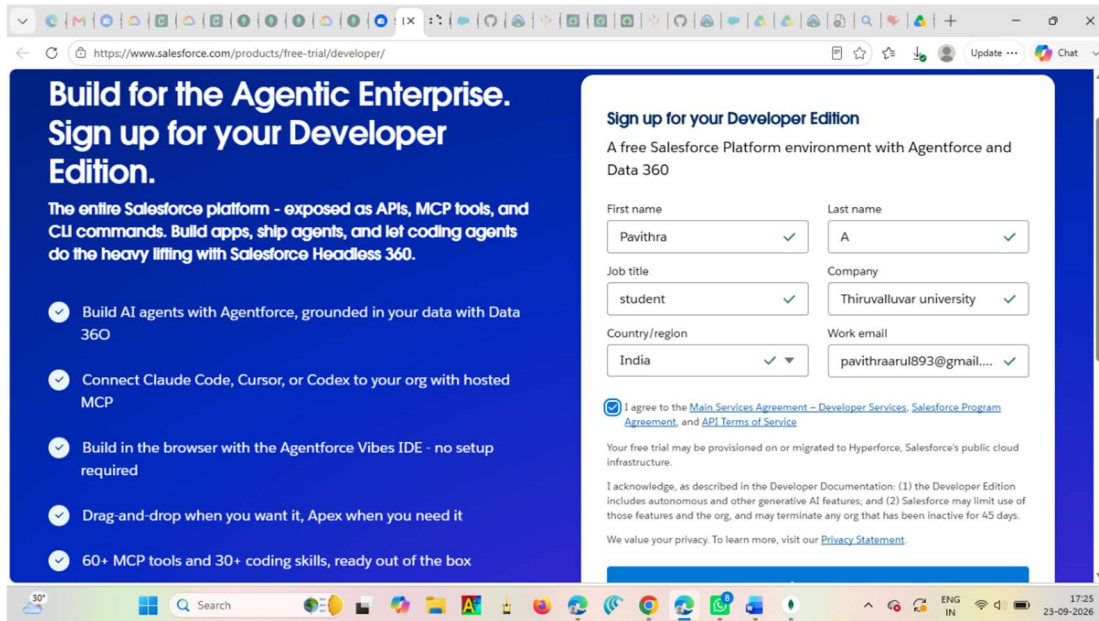
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

<https://youtu.be/r9EX3IGde5k>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to <https://developer.salesforce.com/signup> On the sign-up form, enter the following details :



First name & Last name:

Email:

Role: Developer

Company: College Name County: India

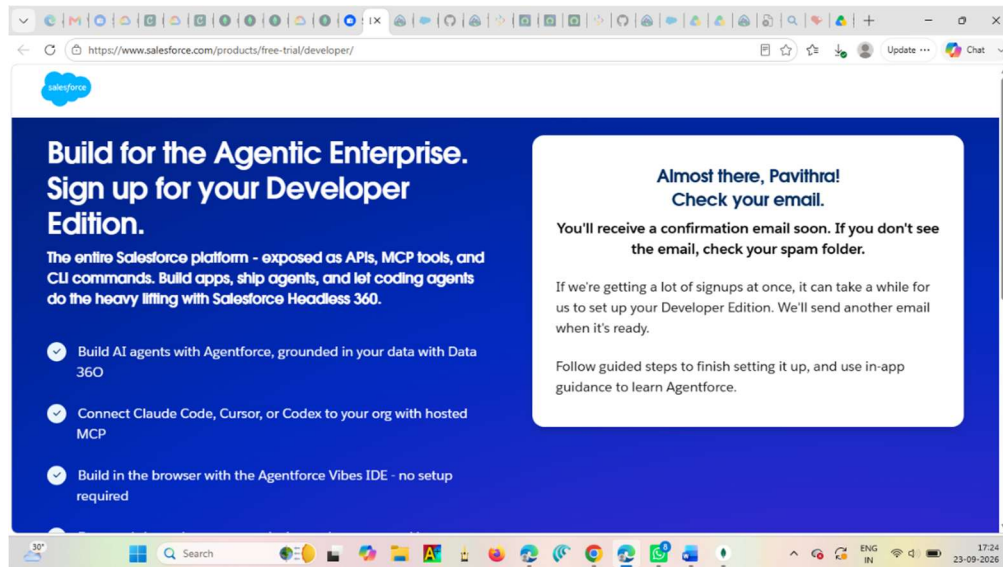
Postal Code: pin code

Username: pavithraarul893@gmail.com

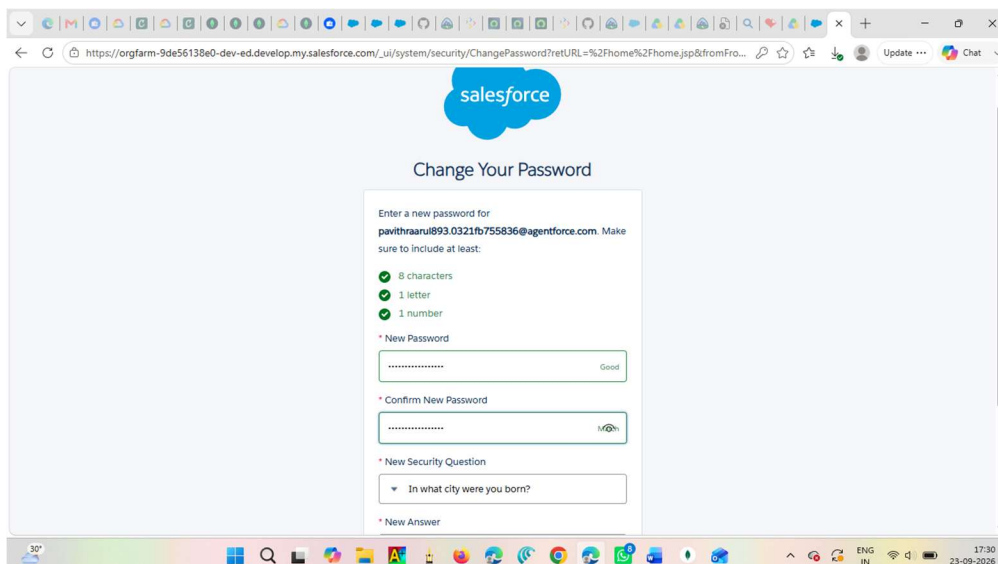
Click on sign me up after filling these.

Activity 2: Account Activation:

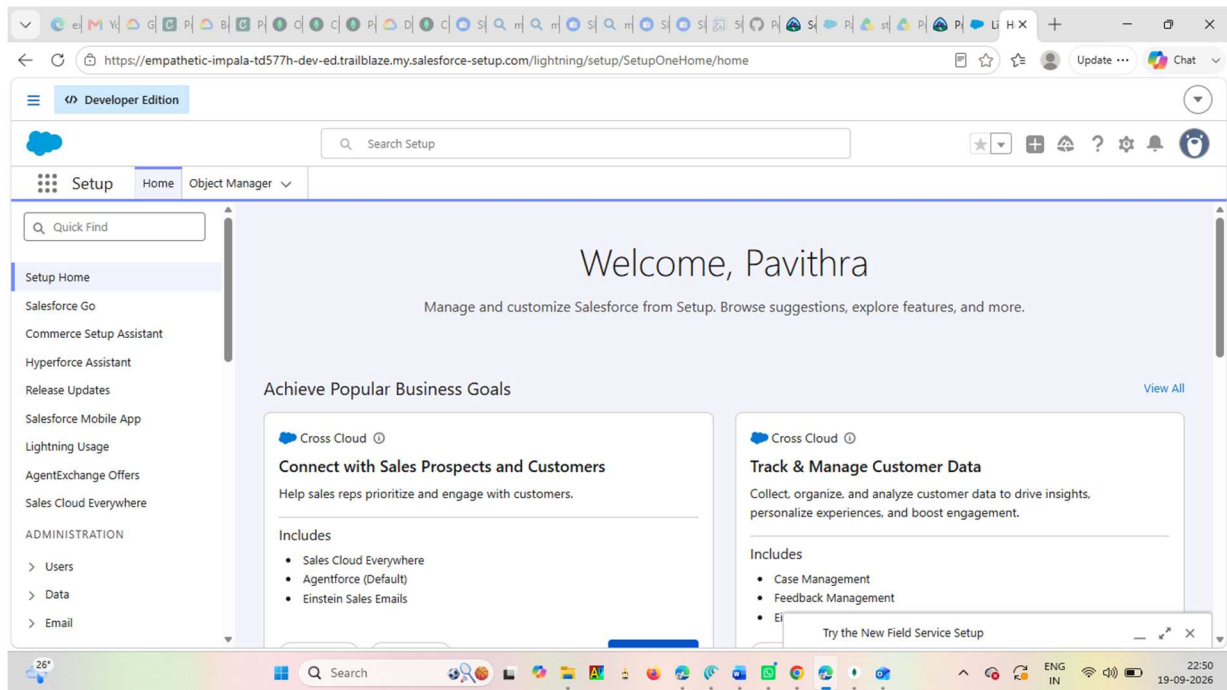
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.



4. Then you will redirect your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the student object:

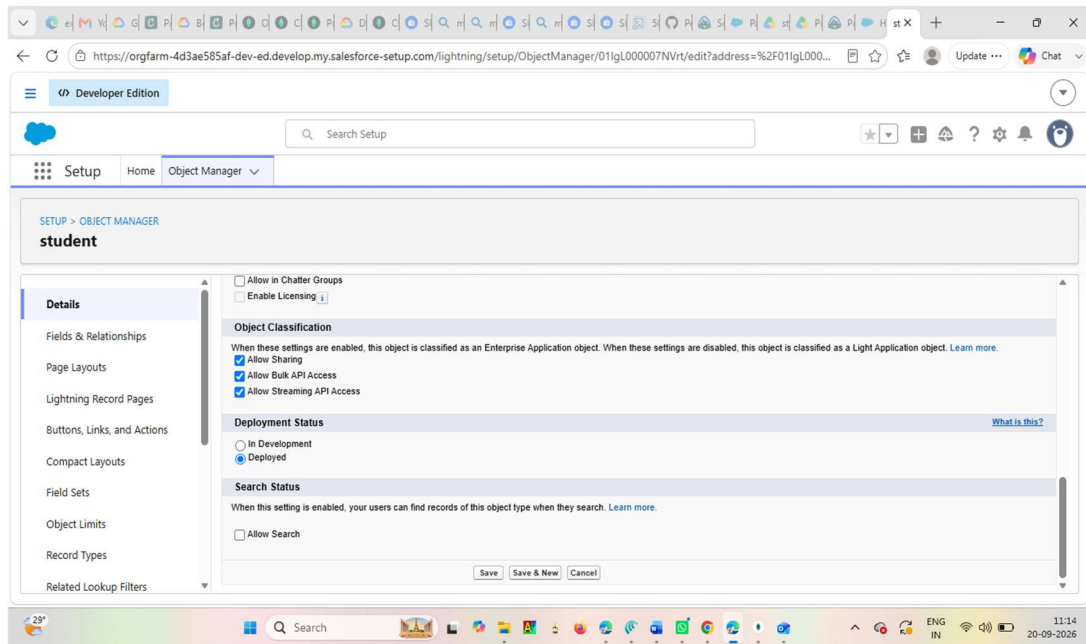
Go to the Setup page. Click on
Object Manager.

Click on Create → Custom Object.

Enter the details below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

The screenshot displays the Salesforce Developer Edition interface. The browser address bar shows the URL: <https://orgfarm-4d3ae585af-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01gl000007NVrt/edit?address=%2F01gl000...>. The page title is "Developer Edition". The navigation bar includes "Setup", "Home", and "Object Manager". The main content area is titled "SETUP > OBJECT MANAGER" and "student". On the left, a sidebar lists various setup options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The "Details" section is selected, showing the "Custom Object Definition Edit" form. The form includes fields for "Label" (student), "Plural Label" (students), "Object Name" (student), and a "Description" text area. There are also checkboxes for "Starts with vowel sound" and "Context-Sensitive Help Setting". The form has "Save", "Save & New", and "Cancel" buttons. The bottom of the screen shows the Windows taskbar with the date and time: 11:13, 20-09-2026.



In the same way Create **Courses, Instructors, and Enrolments** objects

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

Activity 1: Create Lookup from Enrollment to Student:

- Go to Setup → click Object Manager.
- Type Enrollment in Quick Find → click Enrollment object.
- Click Fields & Relationships → click New.
- Select Lookup Relationship → click Next.
- For Related To, select Student c click Next.

Enter Field Label: Student (API Name: Student_____c).

Click Next through Field-Level Security and Layout options (ensure its visible and added to layout). Click Save.

The screenshot shows the 'Object Manager' interface for the 'Student' object. It is at 'Step 1: Choose the field type'. The 'Data Type' section has three options: 'None Selected', 'Lookup Relationship' (which is selected), and 'Master-Detail Relationship'. To the right of the 'Lookup Relationship' option, there is explanatory text: 'Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list. Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: • The relationship field is required on all detail records. • The ownership and sharing of a detail record are determined by the master record.'

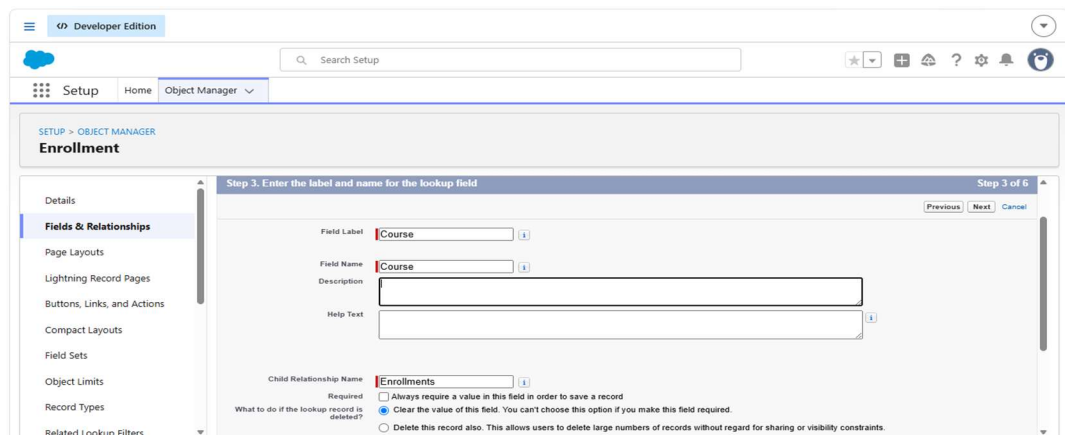
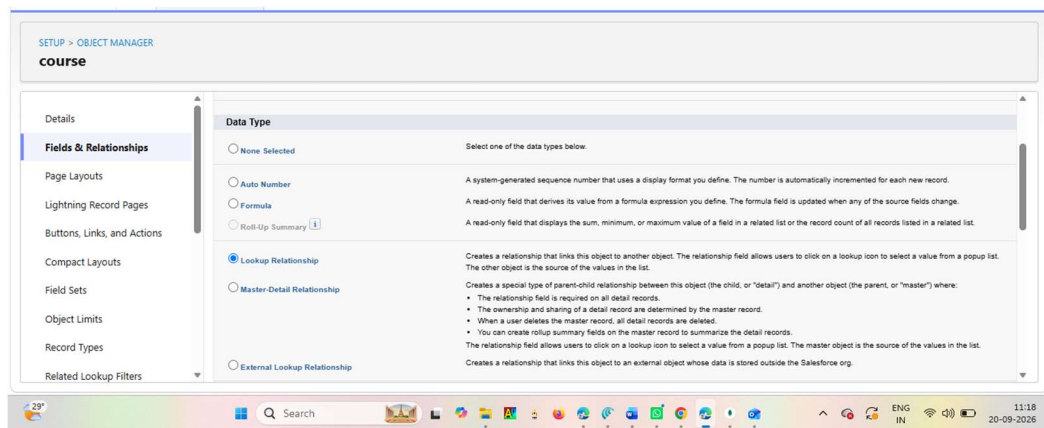
Activity 2: Create Lookup from Enrollment to Course

Purpose: To link each Enrollment to a Course.

The screenshot shows the 'Object Manager' interface for the 'Enrollment' object, specifically the 'Fields & Relationships' tab. The 'Field Information' section shows a new field being created with the following details: Field Label: 'Student', Field Name: 'Student', Data Type: 'Lookup', and Related To: 'Student'. The 'Lookup Options' section shows the 'Related To' field set to 'Student' and the 'Create Relationship Name' field set to 'Enrollments'. There are checkboxes for 'Required' (checked), 'Clear the value of this field. You can't choose this option if you make this field required.', 'Delete the record also. This allows users to delete large numbers of records without regard for sharing or visibility constraints.', and 'Don't allow deletion of the lookup record that's part of a lookup relationship.'

Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Coursec click Next.
5. Field Label: Course (API Name: Course_____c).
6. Click Next (set visibility and layout) → Save.



Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:

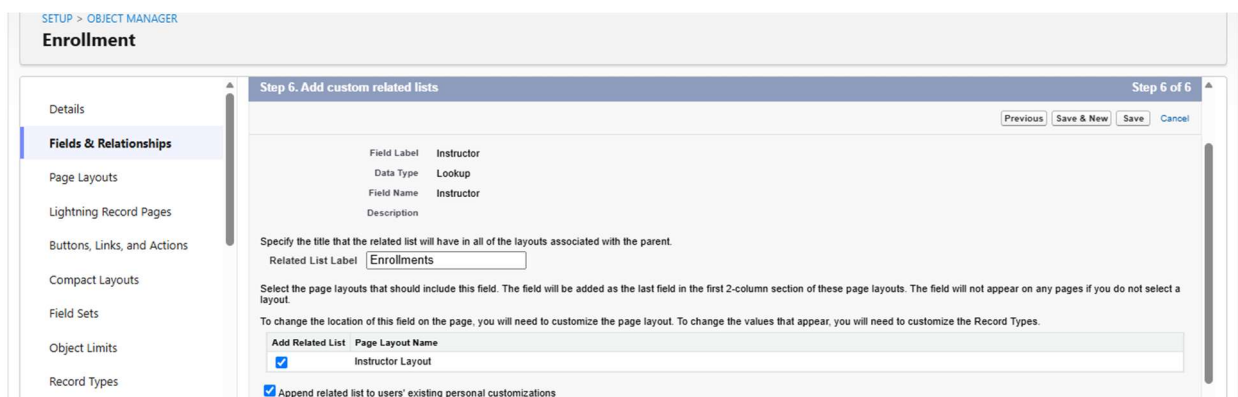
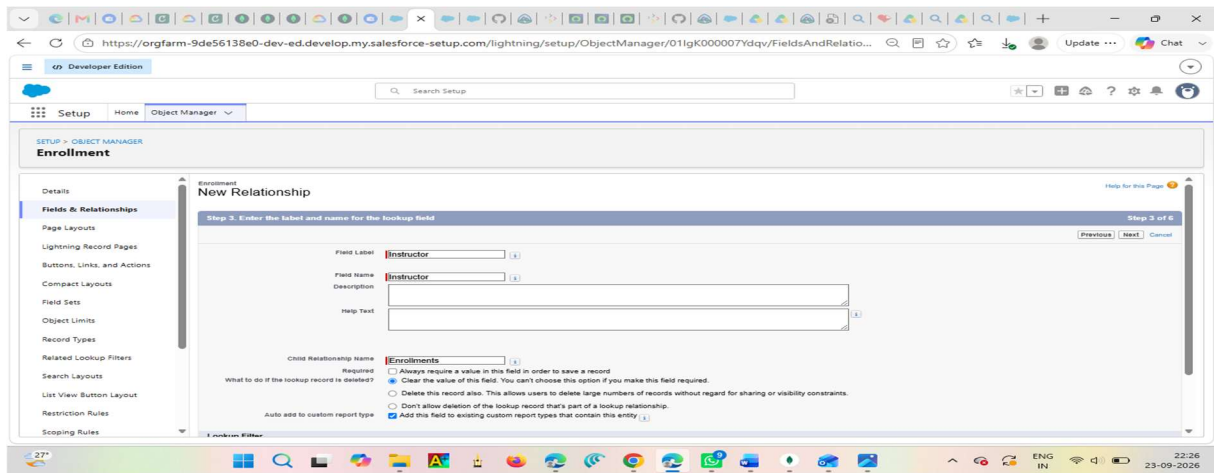
To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor_c Next.
5. Field Label: Instructor (API Name: Instructor_____c).
6. Complete Field-Level Security and Page Layout options → Save

The screenshot shows the Salesforce Setup interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar lists various setup options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, and Object Limits. The main content area is titled 'Enrollment' and shows the 'Fields & Relationships' section. It prompts the user to 'Specify the type of information that the custom field will contain.' Below this, there are several data type options: 'None Selected', 'Auto Number', 'Formula', 'Roll-Up Summary', and 'Lookup Relationship' (which is selected). Each option has a brief description. The 'Lookup Relationship' option is highlighted with a blue circle and a checkmark.

The screenshot shows the 'New Relationship' step in the Salesforce Setup interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar lists various setup options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, and Object Limits. The main content area is titled 'Enrollment' and shows the 'New Relationship' section. It prompts the user to 'Select the other object to which this object is related.' Below this, there is a dropdown menu labeled 'Related To' with 'Instructor' selected. The 'Instructor' text is highlighted in red. The 'Next' button is visible in the top right corner.



Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

Purpose:

To ensure students use only allowed institutional email addresses. Steps:

Go to Setup → Object Manager → Student _____ c.

Click Validation Rules → New.

Rule Name: Student_Email_Domain_Validation. Error Condition

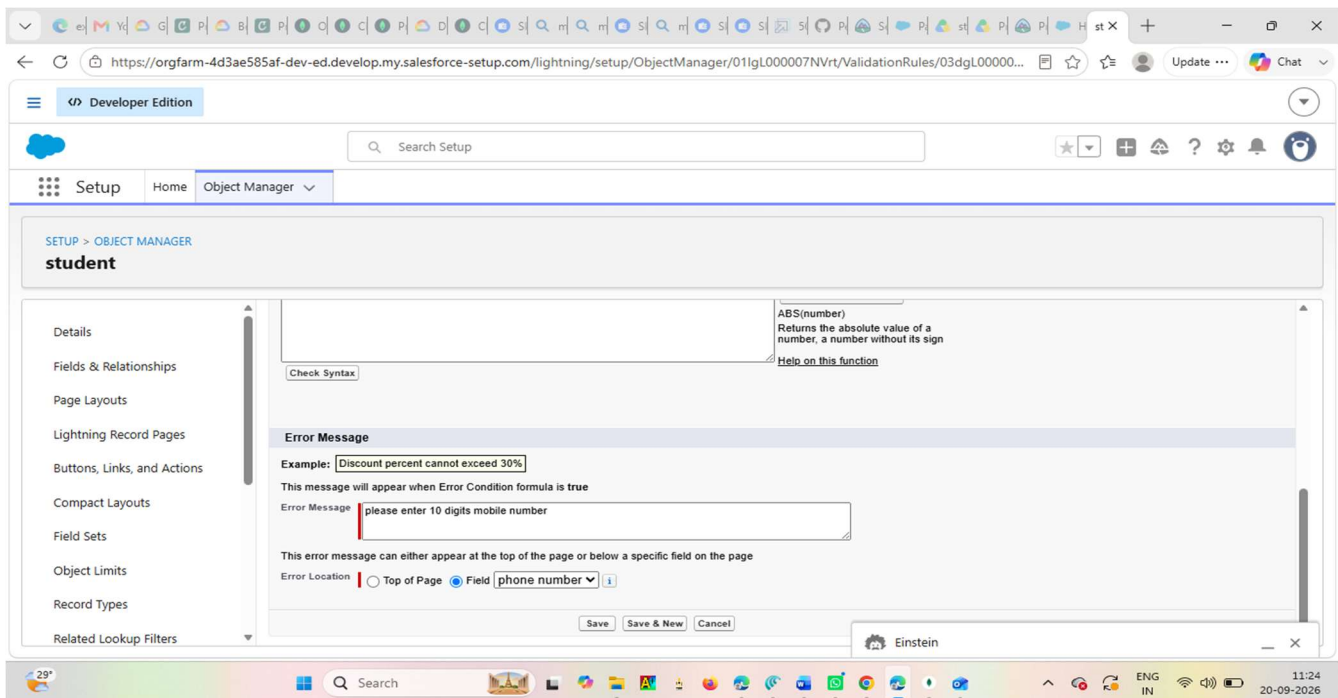
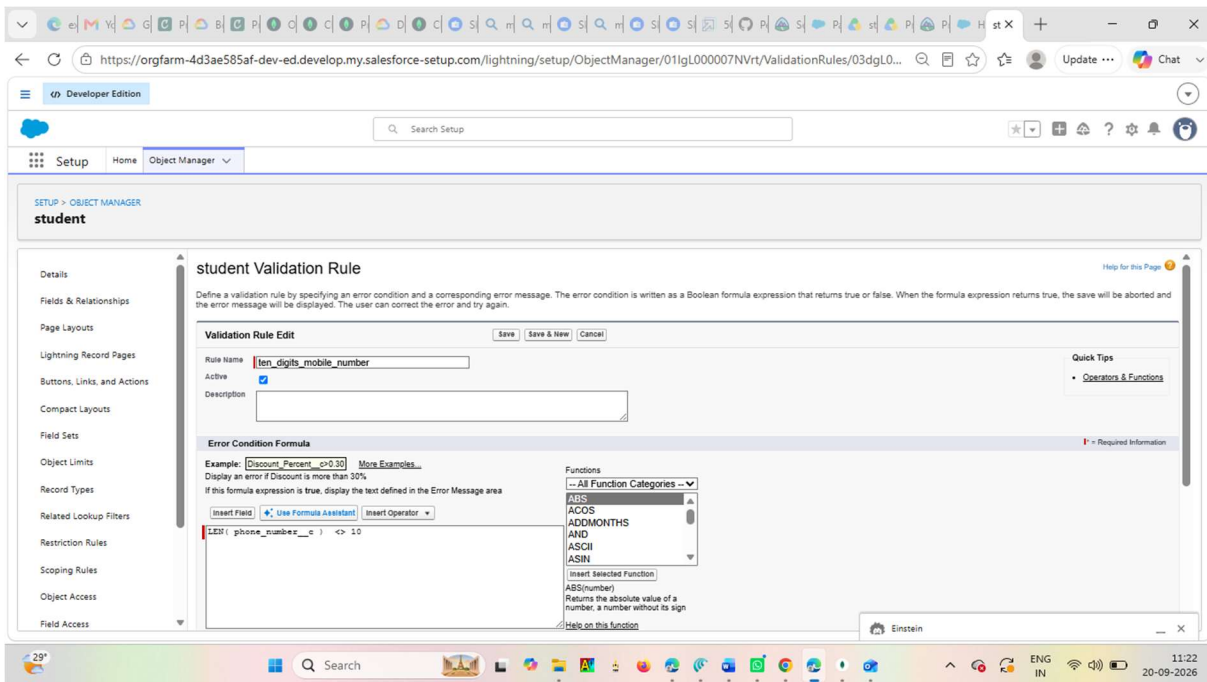
Formula: (example for student.edu)

NOT (CONTAINS(Email_____c, "@student.edu"))

Error Message: Email must be a valid student institutional email. Error Location:

Field → Email_____c.

Click Save.



Activity 2: Fees Must Be Paid Before Approval

Purpose:

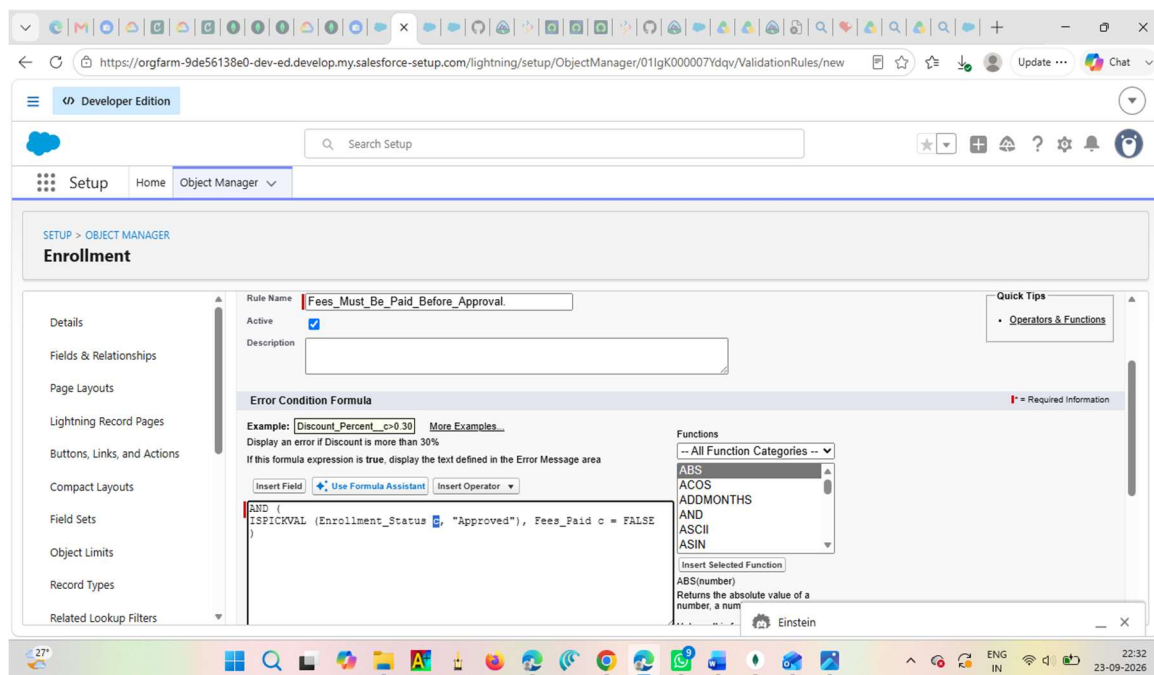
Prevent Enrollment from being approved until fees are paid. Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:

```
AND (  
    ISPICKVAL (Enrollment_Status_____c, "Approved"),  
    Fees_Paid_____c = FALSE  
)
```

5. Error Message: Fees must be paid before approving an enrollment.

6. Error Location: Top of Page (or field as needed). Click Save.



The screenshot shows the Salesforce Setup interface. At the top, there are tabs for Setup, Home, and Object Manager. Below the tabs, the breadcrumb trail reads 'SETUP > OBJECT MANAGER' and the page title is 'Enrollment'. On the left sidebar, there are links for Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, and Compact Layouts. The main content area is titled 'Error Message' and contains the following text: 'Example: Discount percent cannot exceed 30%', 'This message will appear when Error Condition formula is true', and 'Error Message: Fees must be paid before approving an enrollment'. Below this, there is a section for 'Error Location' with two radio buttons: 'Top of Page' (selected) and 'Field'. At the bottom right, there are three buttons: 'Save', 'Save & New', and 'Cancel'.

Milestone 6 – Approval Process

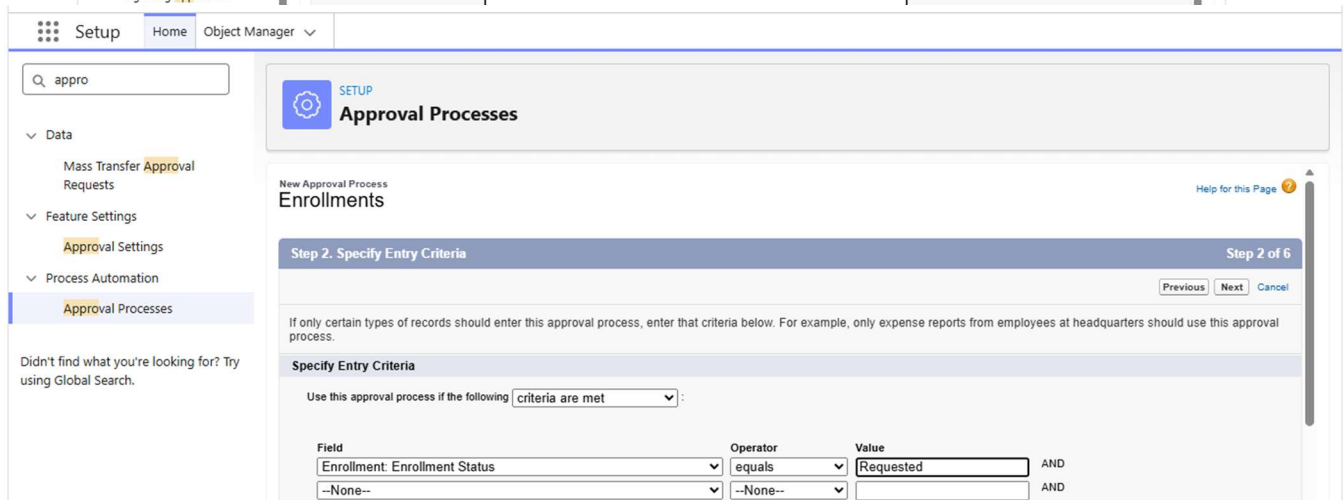
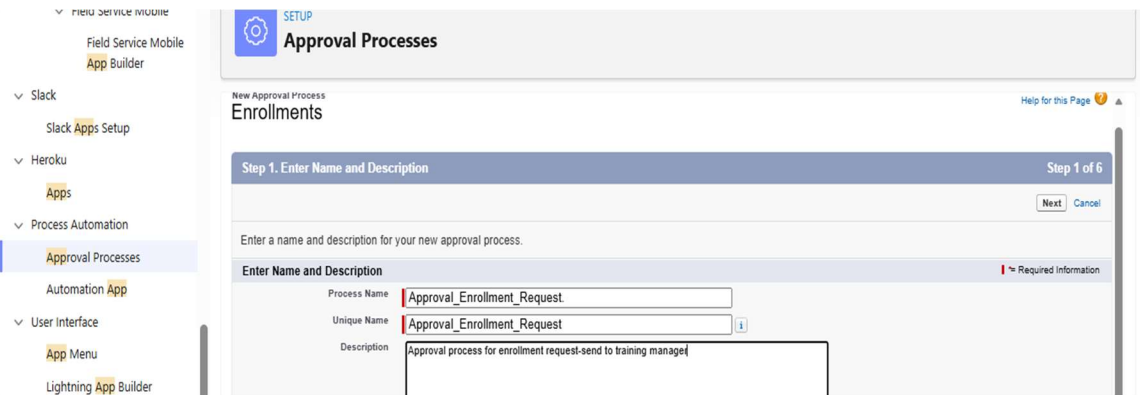
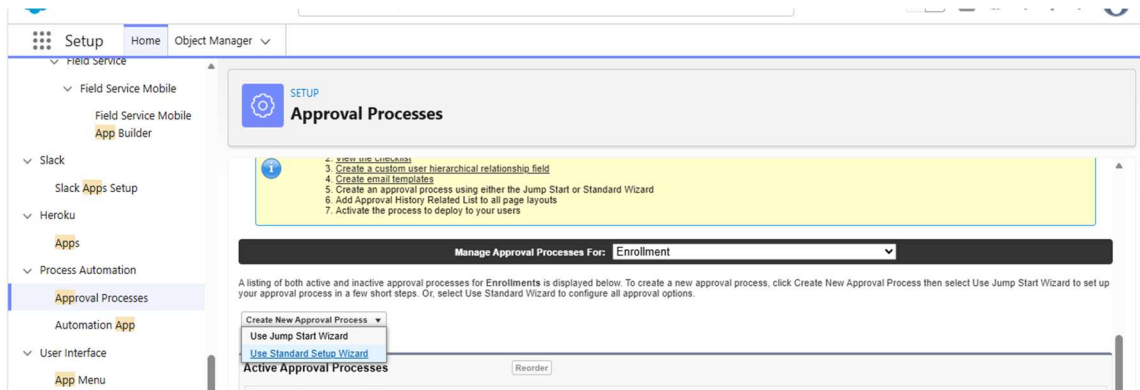
Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Route Enrollment requests to Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment_____c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status_____c = "Requested".
5. Save and configure steps.



6.Add Approval Step:

7.Approver: specific Training Manager user.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'appro' and a list of categories: Data, Feature Settings, Process Automation, and Approval Processes. The 'Approval Processes' category is selected. The main content area is titled 'Approval Processes' and shows a 'New Approval Step' wizard. The first step is 'Enter Name and Description'. The form fields are: Approval Process Name (Approval_Enrollment_Request), Name (Training manager), Unique Name (Training_manager), and Description (empty). A 'Next' button is visible.

1. Under Final Approval Actions:

2. Add Field Update to set Enrollment_Status_____c = "Approved".

The screenshot shows the Salesforce Setup interface for 'Field Updates'. The left sidebar is the same as the previous screenshot. The main content area is titled 'Field Updates' and shows a 'New Field Update' wizard. The first step is 'Field Update Edit'. The form fields are: Name (Status to approved), Unique Name (Status_to_approved), Description (empty), Object (Enrollment), and Field to Update (None--). A 'Next' button is visible.

3. Under Final Rejection Actions:

4. Add Field Update to set Enrollment_Status_____c = "Rejected".

Search Setup

Setup Home Object Manager

Q: appro

▼ Data

Mass Transfer Approval Requests

▼ Feature Settings

Approval Settings

▼ Process Automation

Approval Processes

Didn't find what you're looking for? Try using Global Search.

Field Updates

Set Status to Rejected

Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the field to update may be on a related object. Fields are shown only for the type that you select.

Field Update Edit Save Save & New Cancel

Identification Required Information

Name Set Status to Rejected

Unique Name Set_Status_to_Rejected

Description

Object Enrollment

Field to Update Enrollment Status

Field Data Type Text

Re-evaluate Workflow Rules after Field Change

Specify New Field Value

5. Add Email Alert to send rejection email to students.

Mass Transfer Approval Requests

▼ Feature Settings

Approval Settings

▼ Process Automation

Approval Processes

Didn't find what you're looking for? Try using Global Search.

Edit Email Alert

Required Information

Description Enrollment Rejected

Unique Name Enrollment_Rejected

Object Enrollment

Email Template Enrollment Rejected

Protected Component

Recipient Type Search: User for Find

Recipients

Available Recipients

User: Integration User

User: OrgFarm EPIC

User: Security User

Add

Selected Recipients

User: Pavithra A

You can enter up to five (5) email addresses to be notified.

Additional Emails

pavithraaru893@gmail.com

From Email Address Current User's email address

☐ Make this address the default From email address for this object's email alerts.

Save Save & New Cancel

Email Alerts

6. Activate the approval process.

7. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

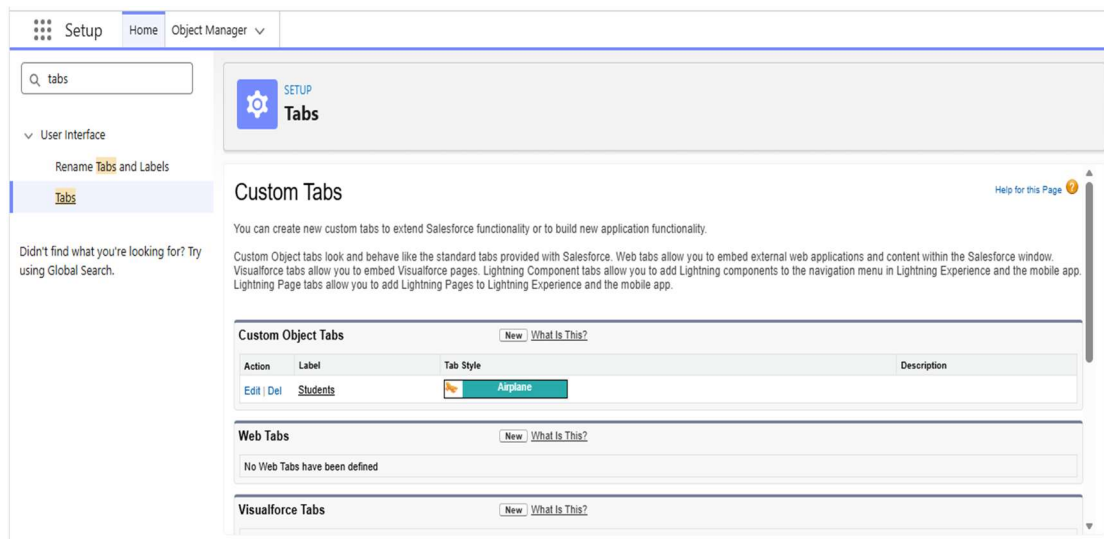
Go to Setup.

In the Quick Find box, type Tabs.

Under Custom Object Tabs, click New.

Select Student from the object list.

1. Choose a tab style (icon).
2. Click Next, assign it to all profiles, and then click Save.



Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.

Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box. Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management Developer Name auto-fills

Description: Optional, but can describe the purpose of the app Click Next and choose the following settings:

App Branding (optional): Add logo or select color theme Navigation Style: Choose Standard Navigation

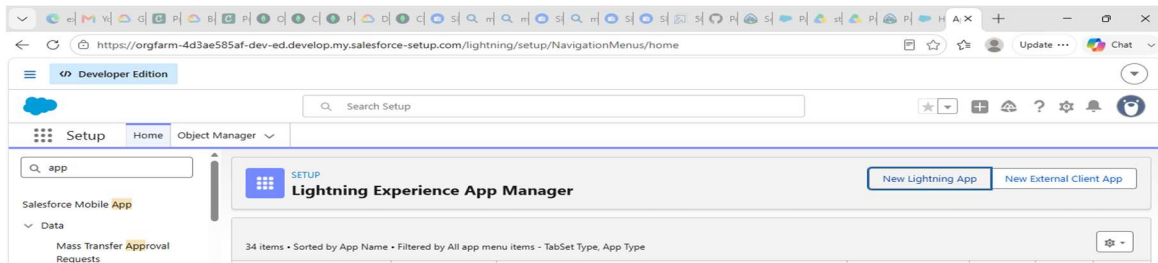
Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.

Finally, click Save & Finish.



App Settings

- App Details & Branding**
- App Options
- Utility Items (Desktop Only)
- Navigation Items
- User Profiles

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ¹
Student enrollment & course management

* Developer Name ¹
Student_enrollment_course

Description ¹
Enter a description...

App Branding

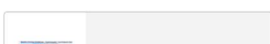
Image ¹

Clear

Primary Color Hex Value ¹
 #0070D2

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview



App Settings

- App Details & Branding
- App Options
- Utility Items (Desktop Only)
- Navigation Items**
- User Profiles

Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the items that you add. Some navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

Available Items

Create

Type to filter list...

- AgentExchange Go-To-Market App Contract Terms
- AgentExchange Go-To-Market App Product Catalogs
- Agentforce Studio
- AI Models
- All Sites
- Alternative Payment Methods
- Analytics

Selected Items

- Students
- Accounts
- Contacts
- Dashboards

App Settings

- App Details & Branding
- App Options
- Utility Items (Desktop Only)
- Navigation Items
- User Profiles**

User Profiles

Choose the user profiles that can access this app.

Available Profiles

Type to filter list...

- Analytics Cloud Integration User
- Analytics Cloud Security User
- Anypoint Integration
- Authenticated Website
- Authenticated Website
- B2B Reordering Portal Buyer Profile
- Contract Manager
- Custom Marketing Profile

Selected Profiles

- System Administrator
- Standard User

Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:

Separating New Enrollment vs Re-Enrollment processes. Steps:

1. Go to Setup → Object Manager → Enrollment_____c.

2. Click Record Types → New.

3. For the first record type:

Existing Record Type: Master

Record Type Label: New Enrollment

Enable for required profiles (Admin, Enrollment Officer).

Select the default Page Layout → Save & New.

4. For the second record type:

Record Type Label: Re-Enrollment

Enable for same profiles.

Select the default Page Layout → Save.

The screenshot shows the Salesforce Setup interface for creating a new record type. The breadcrumb trail is "SETUP > OBJECT MANAGER" and the page title is "Enrollment". A left sidebar contains navigation links: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, and Object Limits. The main content area is titled "Edit Record Type" and "Re-Enrollment". Below this, it says "Enter a new name for the selected record type and click Save." The "Record Type" section contains the following fields: "Record Type Label" with the value "Re-Enrollment", "Record Type Name" with the value "Re_Enrollment", "Namespace Prefix" (empty), and "Description" with the value "used when a student re-enrolls in a course". The "Active" checkbox is checked. A "Required Information" warning icon is visible. At the bottom are "Save" and "Cancel" buttons.

Activity 2: Separate Page Layo Creations for Each Enrollment Record Type

The screenshot shows the Salesforce Setup interface for creating a new record type. The breadcrumb trail is "SETUP > OBJECT MANAGER" and the page title is "Enrollment". A left sidebar contains navigation links: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, and Object Limits. The main content area is titled "Edit Record Type" and "New Enrollment". Below this, it says "Enter a new name for the selected record type and click Save." The "Record Type" section contains the following fields: "Record Type Label" with the value "New Enrollment", "Record Type Name" with the value "New_Enrollment", "Namespace Prefix" (empty), and "Description" with the value "used for first-time enrollments". The "Active" checkbox is checked. A "Required Information" warning icon is visible. At the bottom are "Save" and "Cancel" buttons.

Purpose:

To show different fields for New Enrollment vs Re-Enrollment. Steps:

1. Enrollment _____ c object → click Page Layouts.

2. Click New to create a layout for New Enrollment:

Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.

Keep fields relevant for first-time enrollment: Student, Course, Instructor,

Enrollment Date, Status, Fees, etc.

Remove unnecessary re-enrollment fields if any.

Click Save.

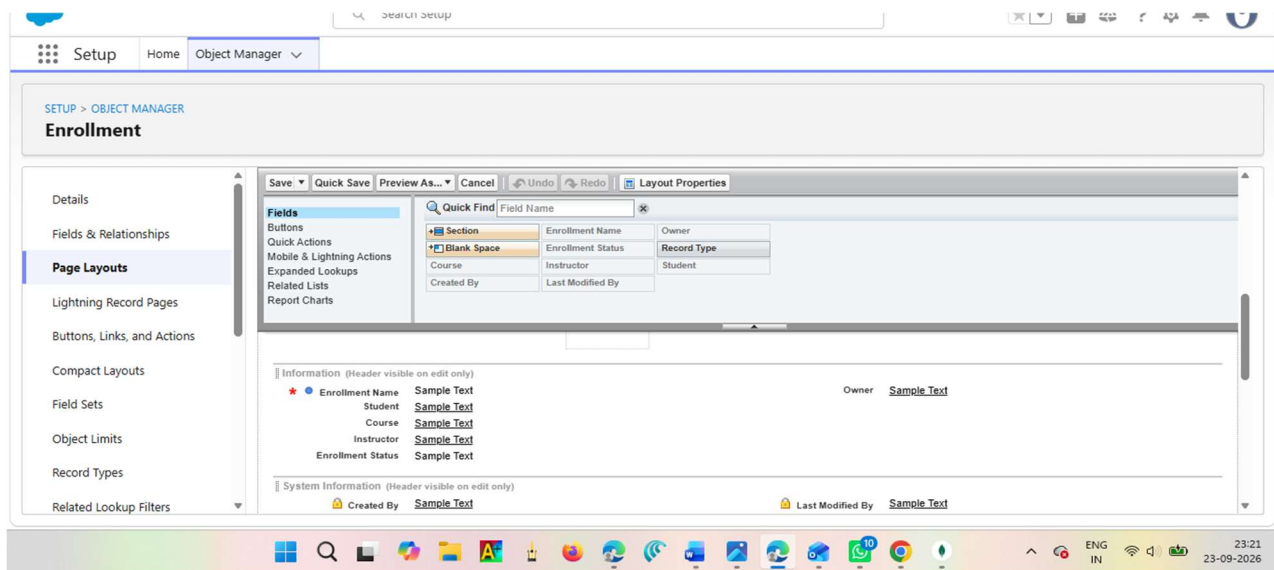
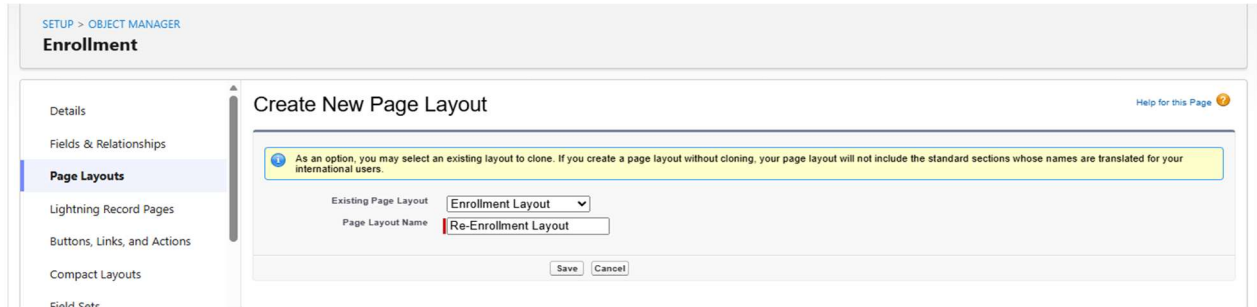
The screenshot shows the 'Create New Page Layout' dialog in Salesforce. The left sidebar is titled 'Enrollment' and has a 'Page Layouts' section highlighted. The main area has a title 'Create New Page Layout' and a help icon. Below the title, there is a message: 'As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard sections whose names are translated for your international users.' Below this message, there are two input fields: 'Existing Page Layout' with a dropdown menu showing 'Enrollment Layout' and 'Page Layout Name' with a text input field containing 'New Enrollment layout'. At the bottom right, there are 'Save' and 'Cancel' buttons.

3. Again, click New to create Re-Enrollment Layout:

The screenshot shows the 'Page Layout' editor in Salesforce. The left sidebar is titled 'Enrollment' and has a 'Page Layouts' section highlighted. The main area has a title 'Page Layout' and a 'Quick Find' search bar. Below the search bar, there is a table of fields with columns 'Field Name' and 'Owner'. The fields are: Enrollment Name, Enrollment Status, Record Type, Course, Instructor, Student, Created By, and Last Modified By. Below the table, there is a section for 'Information (Header visible on edit only)' with fields for Enrollment Name, Student, Course, Instructor, Enrollment Status, and Owner. At the bottom, there is a section for 'System Information (Header visible on edit only)' with fields for Created By and Last Modified By.

Clone existing layout → name its Re-Enrollment Layout → Save.

- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.



4. Click Page Layout Assignment → Edit Assignment.

5. For each profile and record type:

New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout. Click Save.

Activity 3: Enrollment Related Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student_____c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.

Drag Enrollments related list onto the layout. Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course_____c → Page Layouts. Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor_____c → Page Layouts.

Open layout → add Enrollments related list.

Save.

Milestone 10 – Automation with Flow

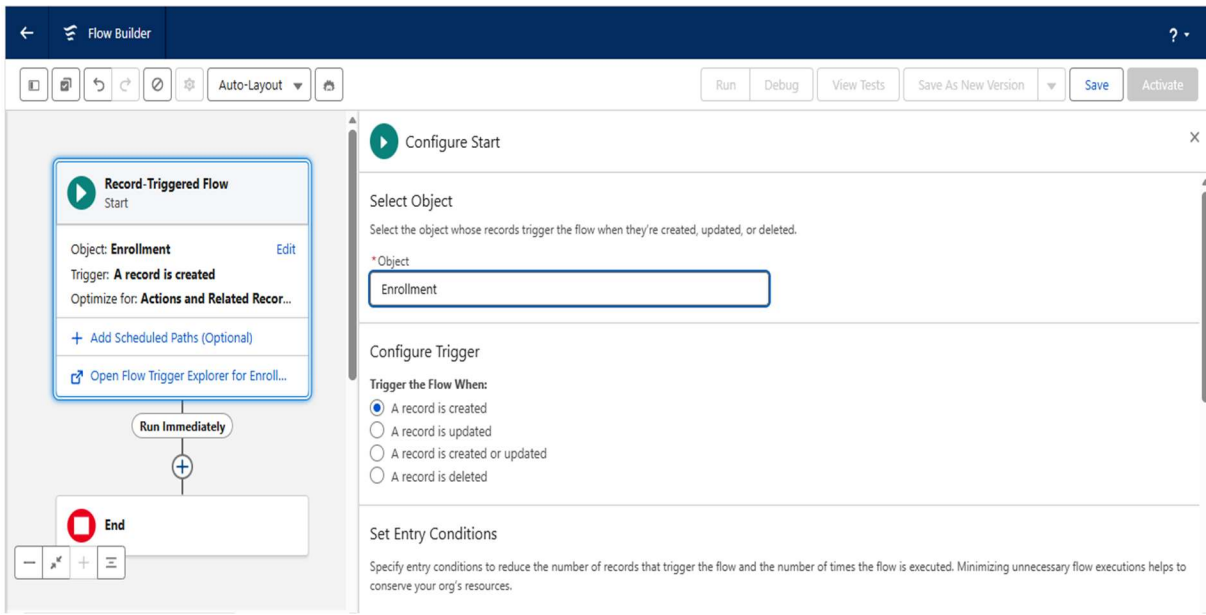
Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment Date_____c to today when Enrollment is created.

Steps:

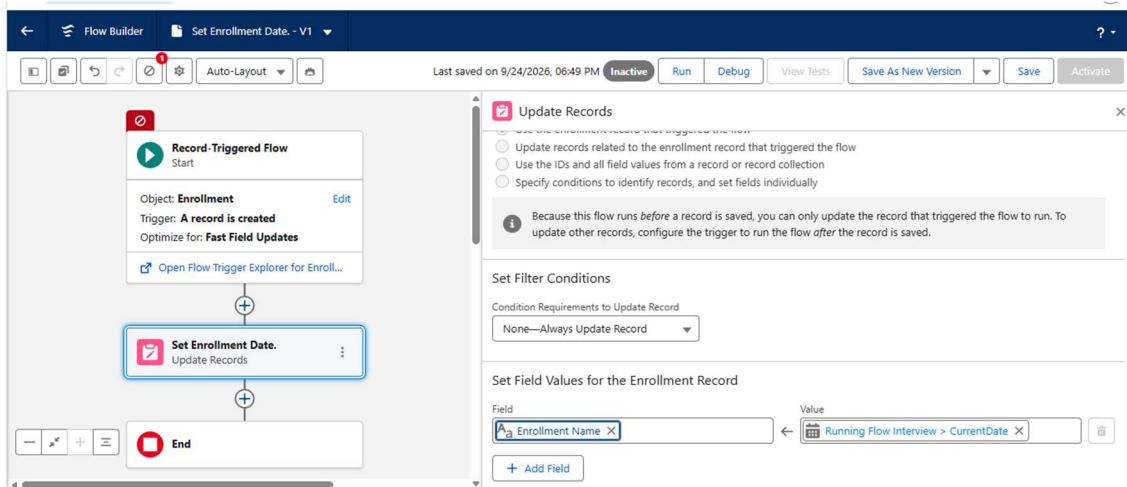
1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment_____c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.



6. Add Update Records element:

- Label: Set Enrollment Date.
- Use record that triggered the flow.
- Field: Enrollment Date_____c → Value: \$Flow.CurrentDate.

7. Save as Set Enrollment Date → Activate.



Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment Status_____c = Approved, send email, alert admin, and set Student to Active.

Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment_____c, Trigger on update.
4. Condition: Enrollment Status_____c Equals "Approved".

Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

- ☐ A record is created
- ☐ A record is updated
- ☒ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

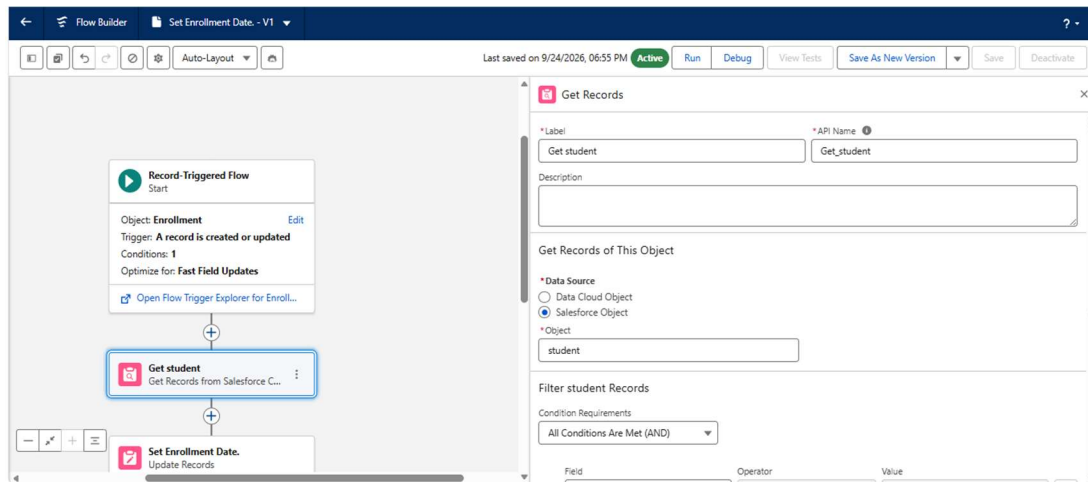
If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Approved

5. Add Get Records to fetch related Student_____c.



6. Add Action (Email Alert) to send email to students.

Enrollment Approved EmailAlert

* Label

* API Name ⓘ

Description

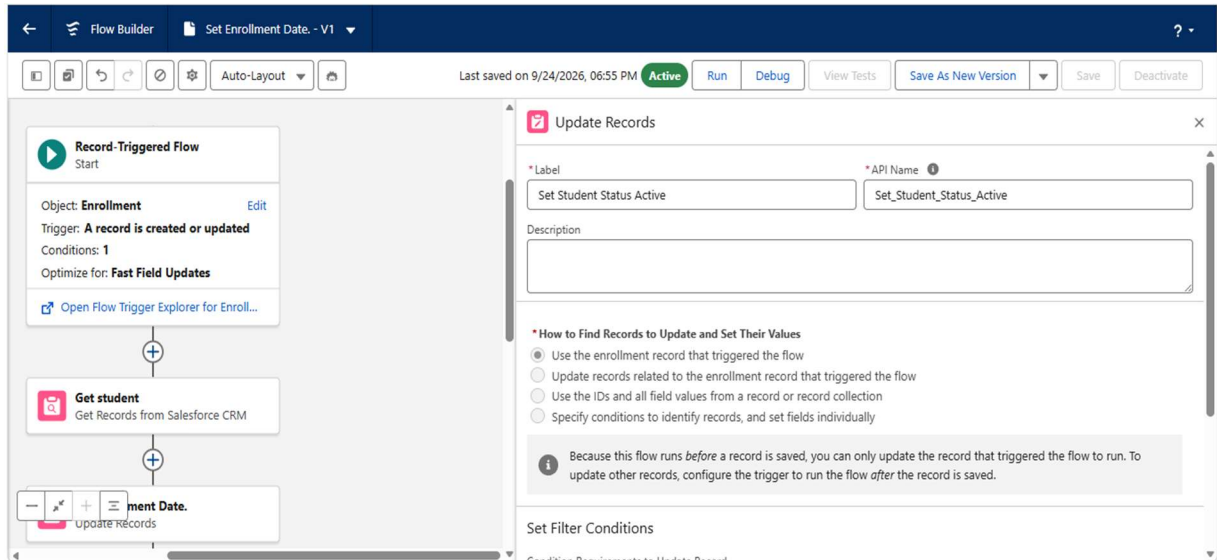
Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

A_a * Record ID ⓘ

7. Add Action to notify Admin if needed.

8. Add Update Records to set Student_____c.Status_____c = "Active".



9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose:

Assign Instructor automatically based on Course category. Rules:

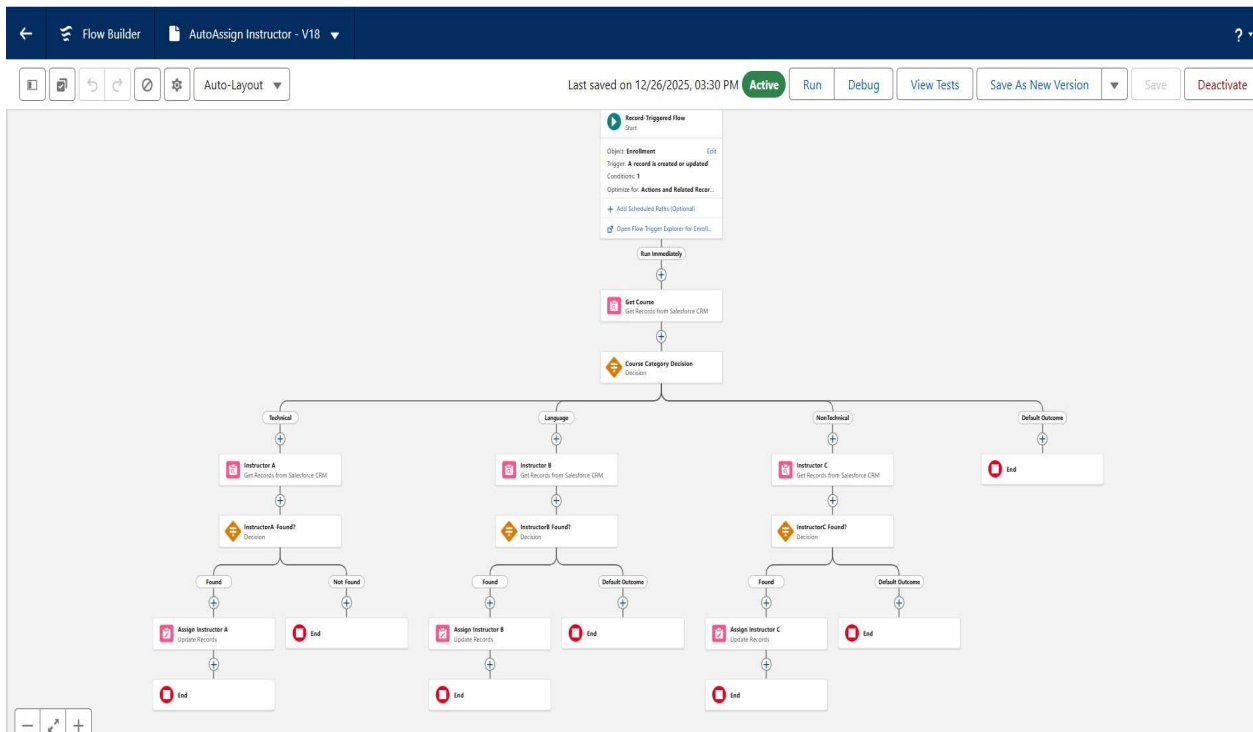
Technical → Instructor A Language

→ Instructor B Non-Technical →

Instructor C

Steps:

1. Prepare Instructor_____c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment_____c (created/updated).
3. Add Get Records to fetch related Course_____c.
4. Add Decision element on Course_____r.Category_____c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor_____c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



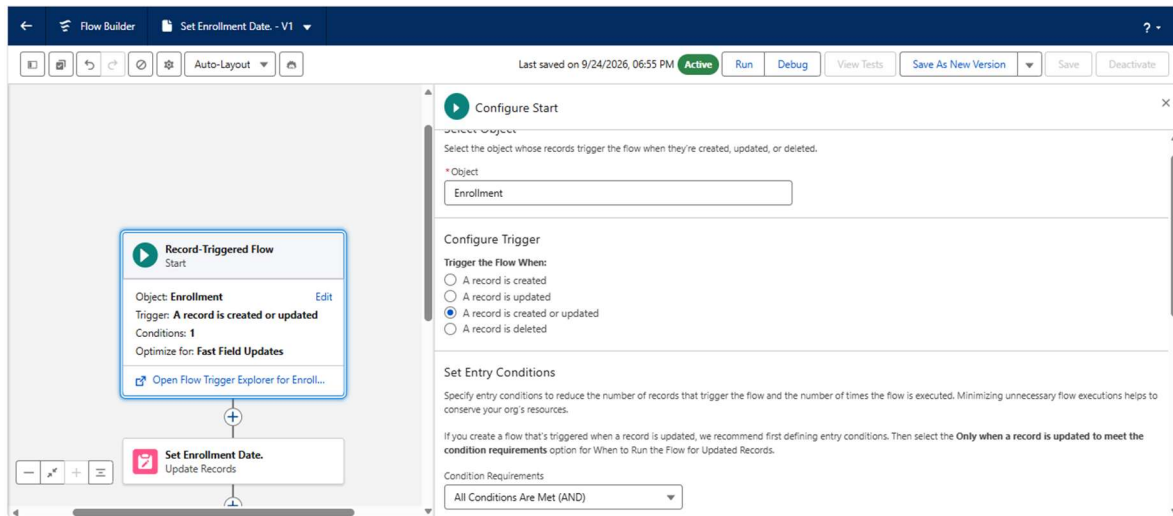
Activity 4: Auto-Create Task for Follow-Up

Purpose:

When Enrollment is “Requested”, create a follow-up task for Admin.

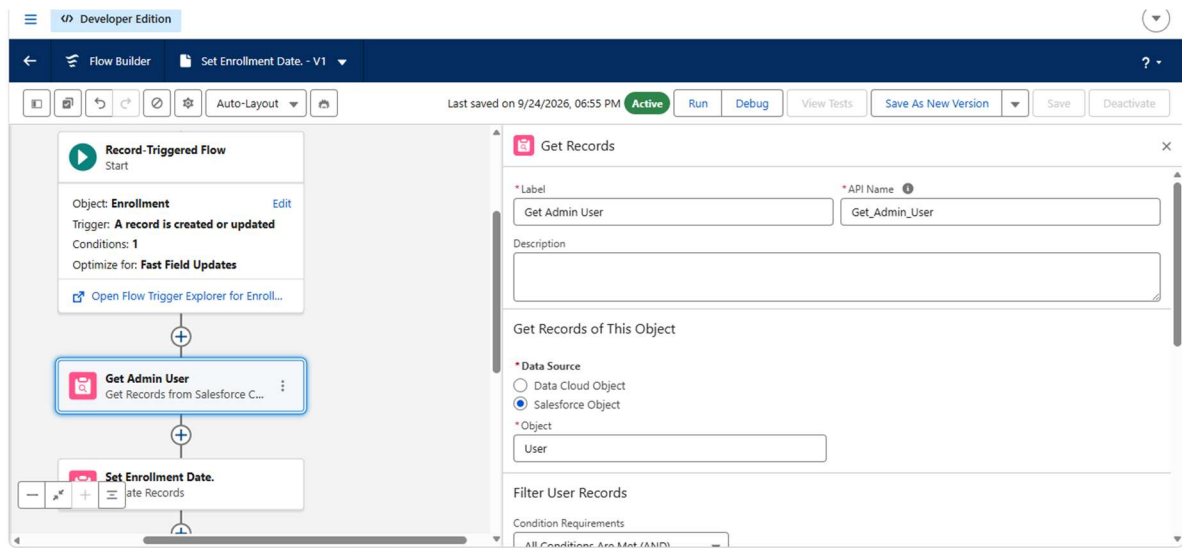
Steps:

1. Create Record-Triggered Flow on Enrollment_____c.
2. Trigger: Record created or updated; condition Enrollment_Status_____c
= "Requested".



3. Add Get Records to fetch Admin user (by Username / Profile).

4. Add Formula (Date): TODAY () + 2 for due date.



5. Add Create Records for Task:

6. Subject: Follow up on enrollment request.

7. Owner Id: Admin user Id.

8. What Id: Enrollment Id.

9. ActivityDate: formula Today + 2.

Create Records

Create a Record of This Object

* Object

Task

Set Field Values for the Task

Field

Due Date Only

Value

DueDateTodayPlus2

Field

Description

Value

Auto-created follow-up for enrollment request

Field

Priority

Value

Normal

Field

Status

Value

Not Started

Field

Subject

Value

Follow up on enrollment request

Field

Related To ID

Value

Triggering Enrolement_c > Record ID

10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

1. Go to **Reports** → **New Report**.
2. Report Type: **Students**.
3. Group rows by **Status_____c**.
4. Add columns as needed (Name, Email, Phone).
5. Save as **Students by Status**.

REPORT ▾

Students by Status ▾ Students

Outline

Filters

Groups

GROUP ROWS

Add group...

Status

GROUP COLUMNS

Add group...

Columns

Add column...

Student: Student Name

Last Name

Phone

Email

Previewing a limited number of records. Run the report to see everything.

Update Preview Automatically

Record Count

0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3 3.2 3.4 3.6 3.8 4

Status

Active

Status: ↑

Student: Student Name

Last Name

Phone

Email

Active (4)

Abhishek Jakkula 12345625 abhishek@student.edu

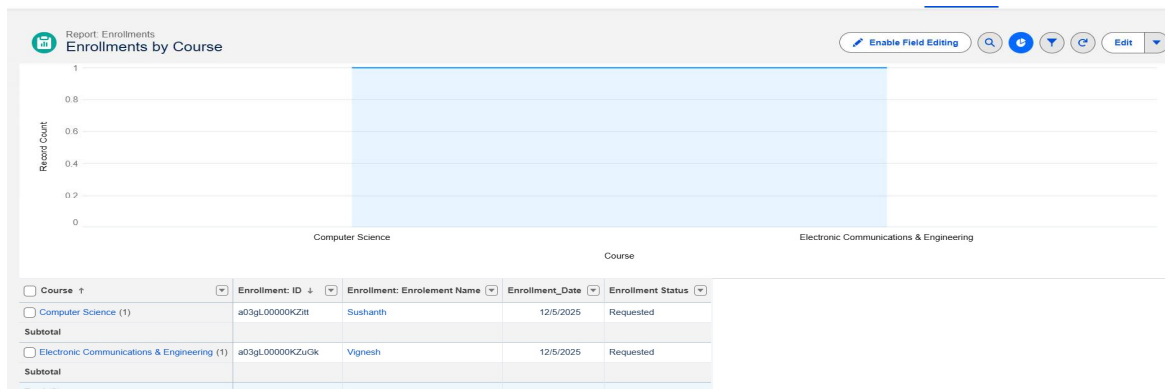
Vignesh Doodh 12345678 vignesh@student.edu

Swaroop M 212154231 swaroop@student.edu

Sushanth jaligama 9000000015 sushanth@student.edu

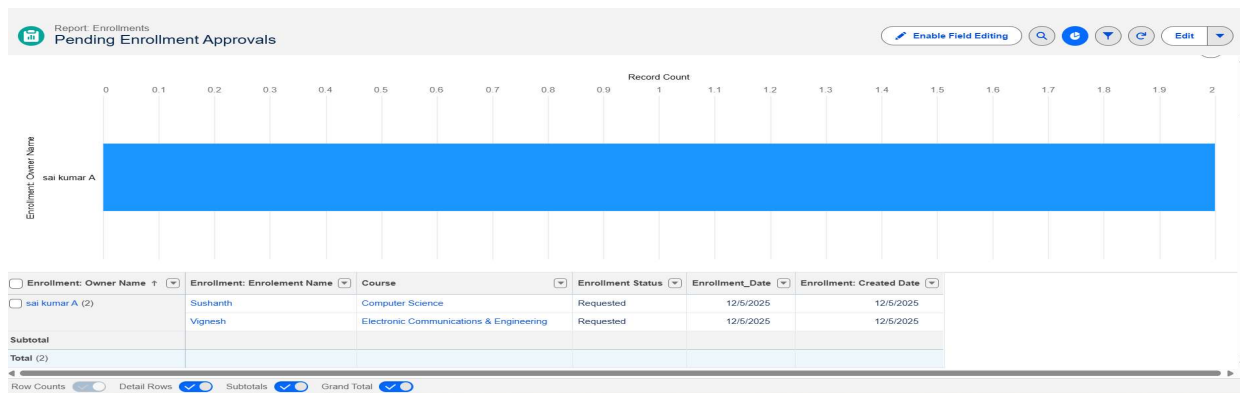
Subtotal

Total (4)



Activity 3: Create “Pending Enrollment Approvals” Report

- 1 . New Report → **Enrollments** report type.
- 2 . Filter: **Enrollment_Status** c = "Requested".
- 3 . Add columns (Student, Course, Owner, Created Date).
- 4 . Save as **Pending Enrollment Approvals**.

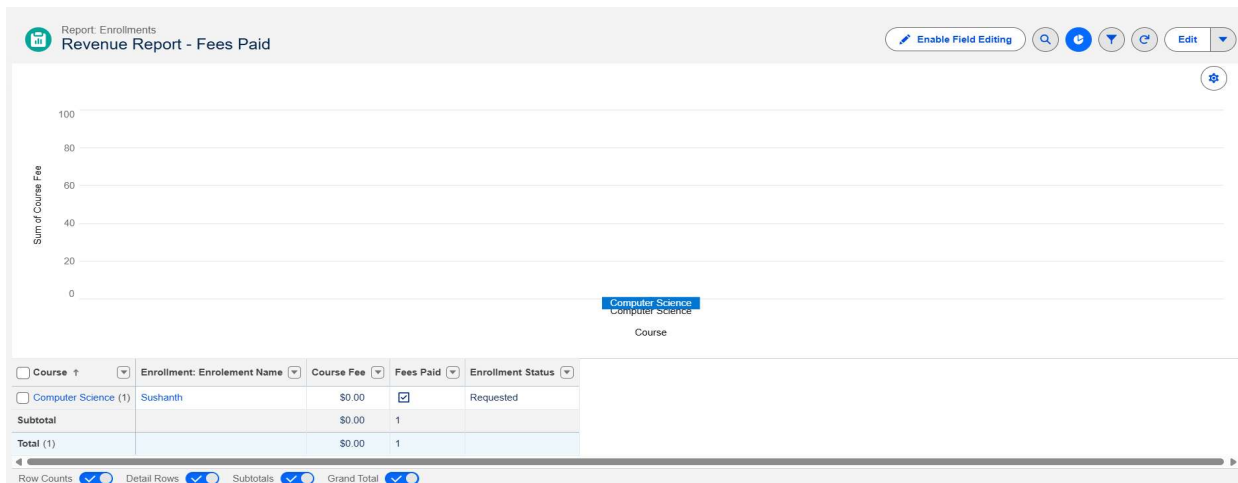


Activity 4: Create “Revenue Report (Fees Paid)” Report

- 1 . New Report → **Enrollments** report type.
- 2 . Filter: **Fees_Paid** c = TRUE.
- 3 . Group by **Course** r.Name.
- 4 . Summarize **Course_Fee** c or similar fee field.
- 5 . Save as **Revenue Report – Fees Paid**.

Activity 4: Create “Revenue Report (Fees Paid)” Report

- 6 . New Report → **Enrollments** report type.
- 7 . Filter: **Fees_Paid**_____c = TRUE.
- 8 . Group by **Course**_____r.Name.
- 9 . Summarize **Course_Fee**_____c or similar fee field
- 10 . Save as **Revenue Report – Fees Paid**.



Milestone 12 – Dashboards:

Activity 1: Create “Student Management Dashboards”

Steps:

- 1 . Go to **Dashboards** → **New Dashboard**.
- 2 . Name: **Student Management Dashboard**.

DASHBOARD NAME	DESCRIPTION	FOLDER	CREATED BY	CREATED ON	SUBSCRIBED
Student Management Dashboard		Global Sales Dashboards	Pavithra A	9/24/2026, 7:22 AM	
Big Deals		Private Dashboards	Pavithra A	8/17/2026, 6:30 AM	
Global Sales Dashboard		Global Sales Dashboards	Pavithra A	8/16/2026, 7:24 AM	
pavithraarul		Private Dashboards	Pavithra A	8/8/2026, 8:05 AM	

- 3 . Select folder → **Create**.
- 4 . Select new Dashboard folder and save.

Add components:

Donut Chart → using **Students by Status** report.

Bar Chart → using **Enrollments by Course** report

Gauge → using **Revenue**
revenue field).

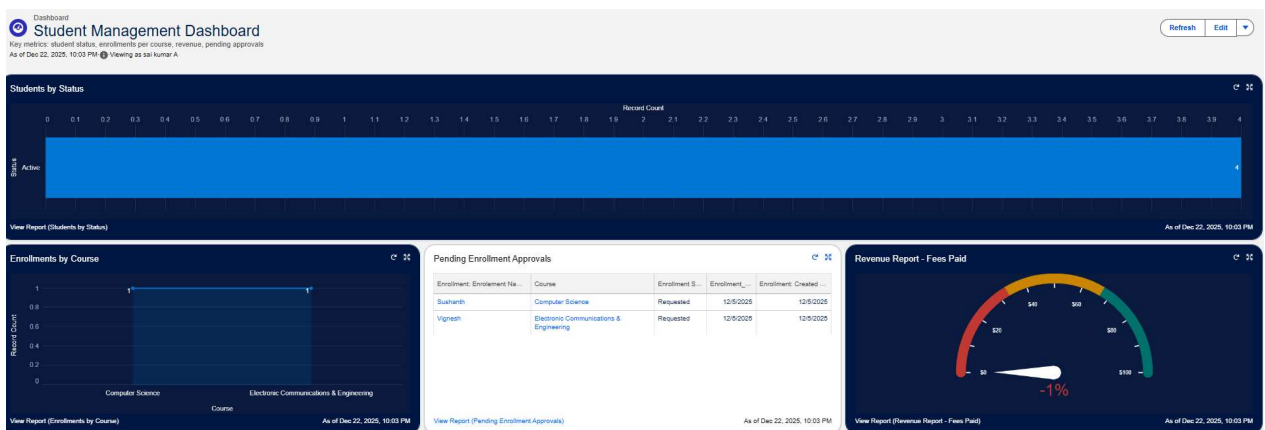
Report – Fees Paid (sum of

Table → using **Pending**

Enrollment Approvals report.

5 . Arrange components on canvas.

6 . Click **Save** → **Done**.



Milestone 13 – Security Setup

Activity 1: Configure Profiles

);

1 . Go to **Setup** → **Profiles** → **New** (or Clone **Profiles Required:**

Admin – Full access.

Enrollment Officer – Create/Update Enrollments.

Instructor – Read-only access to assigned Enrollments.

2 . Steps (example for **Enrollment Officer** fName: **Enrollment Officer**).

3 . In **Object Settings**:

○ **Enrollment**_____c: Read, Create, Edit.

○ **Student**_____c / **Course**_____c: Read (and Edit if required).

4 . Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments.

Activity 2: Create Permission Sets

- 1 . Go to **Setup** → **Permission Sets** → **New**.
- 2 . Create **Custom Object Access** (additional access to project objects if needed).
- 3 . Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).
- 4 . Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

- 1 . Go to **Setup** → **Sharing Settings**.
- 2 . Set **Organization-Wide Defaults**:
Students (Student__c): Public Read/Write.
Courses (Course__c): Public Read/Write.
Enrollments (Enrollment__c): Private.
- 3 . Save.

For Instructor access to assigned Enrollments, optionally:

Add a **User lookup** on Enrollment (Instructor User).

- 4 . Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

7.FUNCTIONAL & PERFORMANCE TESTING

Functional and performance testing for the Student Management & Course Management System (SECMS) ensures that all core features—such as user authentication, student enrollment, course creation and registration, attendance marking, grade entry, and reporting—work correctly and efficiently under real-world conditions. Functional testing uses structured test cases to validate each requirement, confirming that valid and invalid inputs are handled properly, role-based access is enforced, and business rules like prerequisites and seat limits are applied correctly. Performance testing complements this by simulating multiple concurrent users and large data volumes to verify that the system maintains acceptable response times, remains stable during peak enrollment periods, and scales without errors or data corruption, thereby ensuring a reliable and responsive platform for students, faculty, and administrators.

8.ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9.Conclusion:

Deployment activities include validating all configurations, automations, flows, security settings, and data within the Salesforce environment. Final checks are performed to ensure that the application functions correctly without errors and meets the defined business requirements.

The final presentation demonstrates the complete working of the project, showcasing key features such as record creation, automation, approval processes, reports, and dashboards. This phase highlights the practical implementation of Salesforce concepts and the overall value delivered by the solution.

10.FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11.APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You